



NextAIRE

Next Generation AI Researchers for Air Quality Excellence



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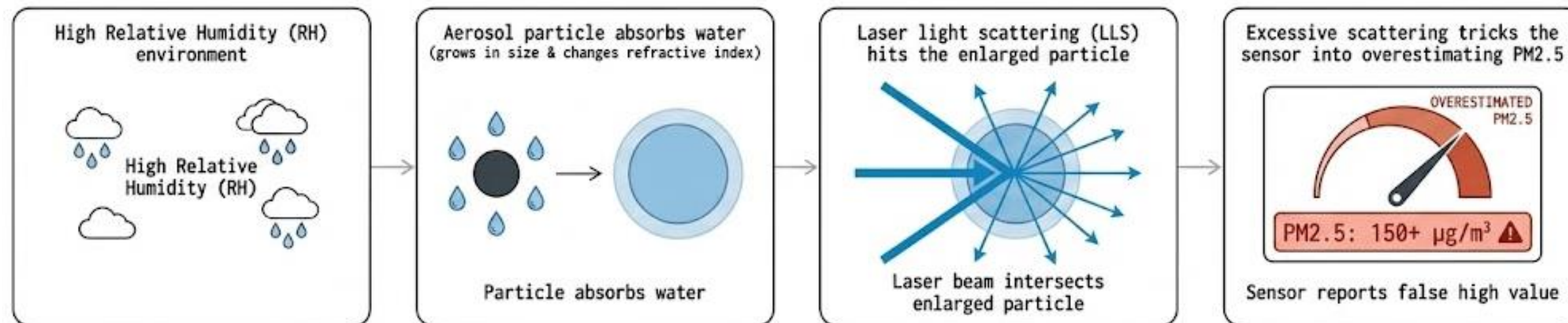
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Calibration of PM sensors

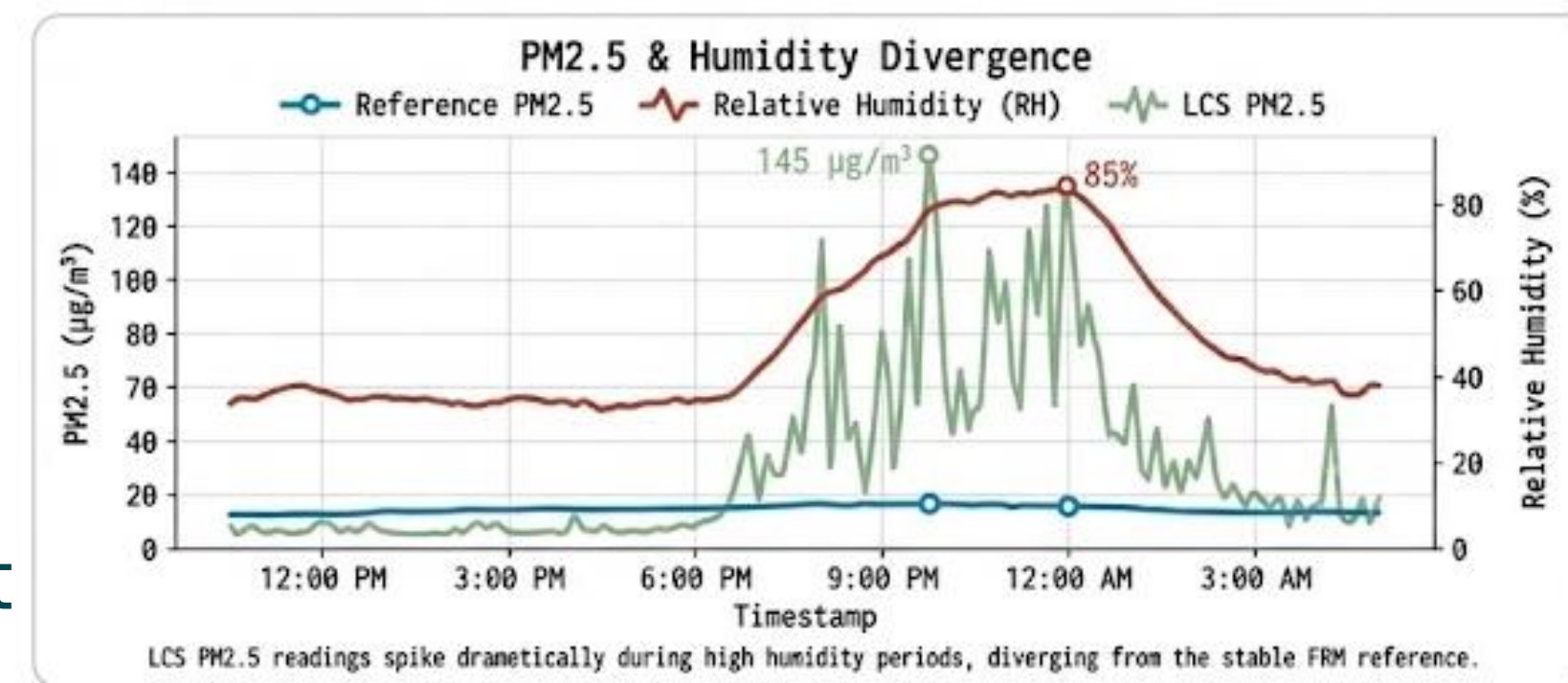
Calibration methods

- Linear regression (default – industry)
- Multiple linear regression
- Orthogonal regression
- Machine learning
- AI models

Weather conditions systematically distort low-cost sensor data

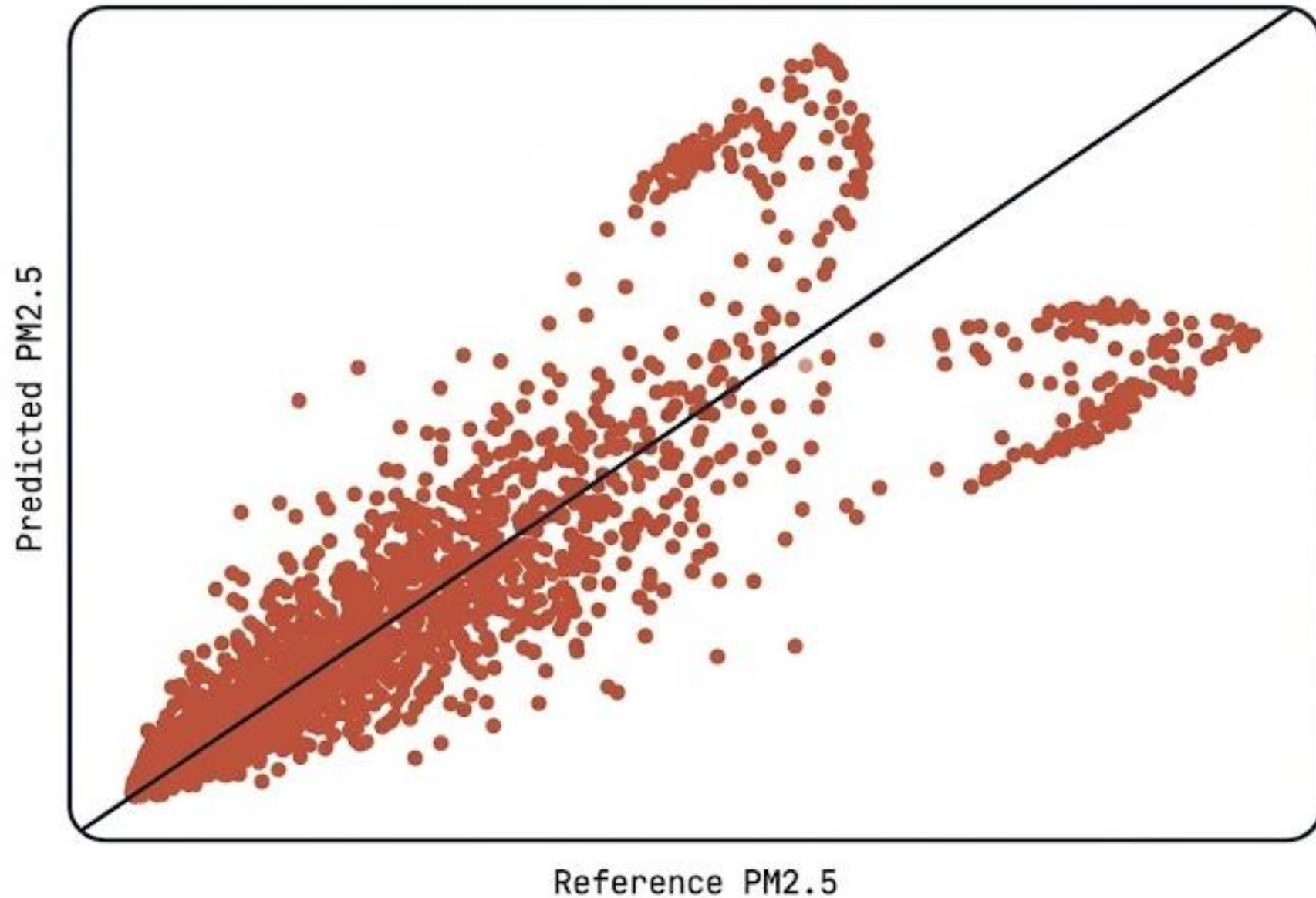


- Key distinction:
 - FRMs utilize mechanical drying systems
 - LCS operates entirely exposed to ambient conditions



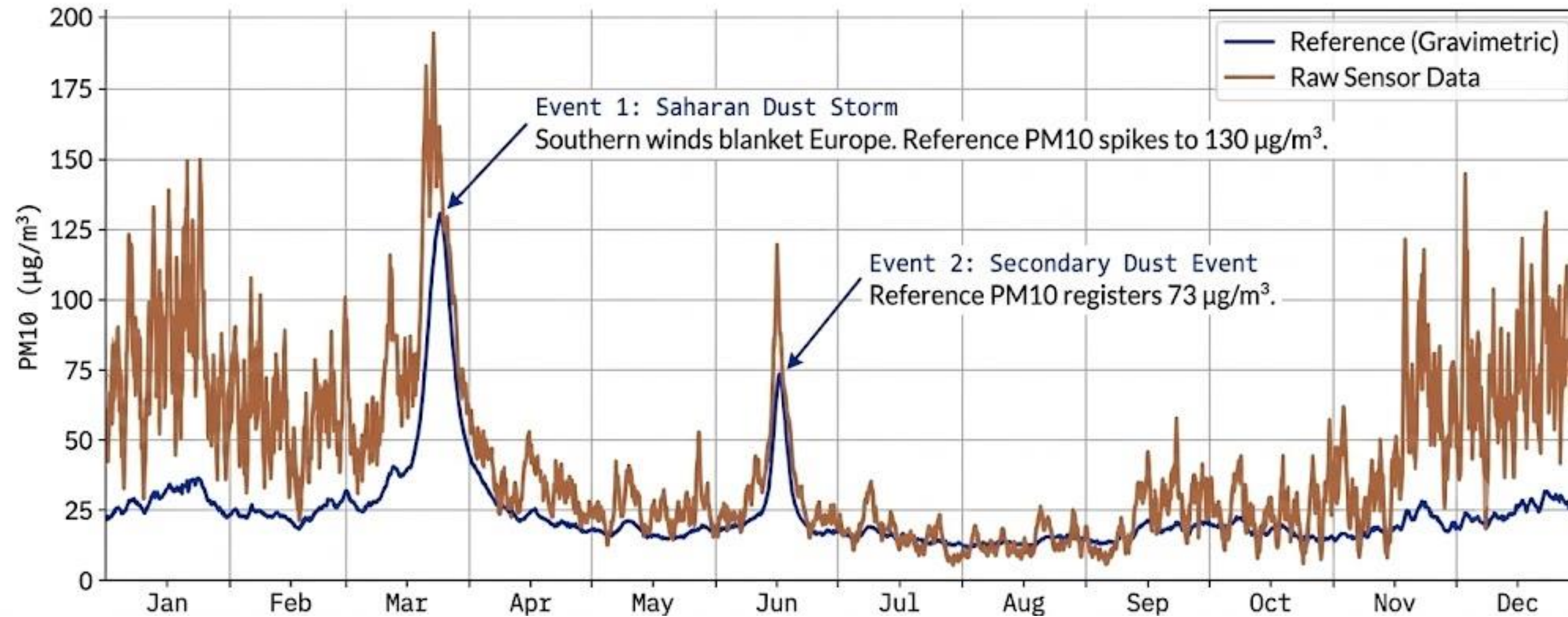
Linear models

(fail to capture atmospheric complexity)



- **Relationship between ambient weather and particulate scattering is highly non-linear**

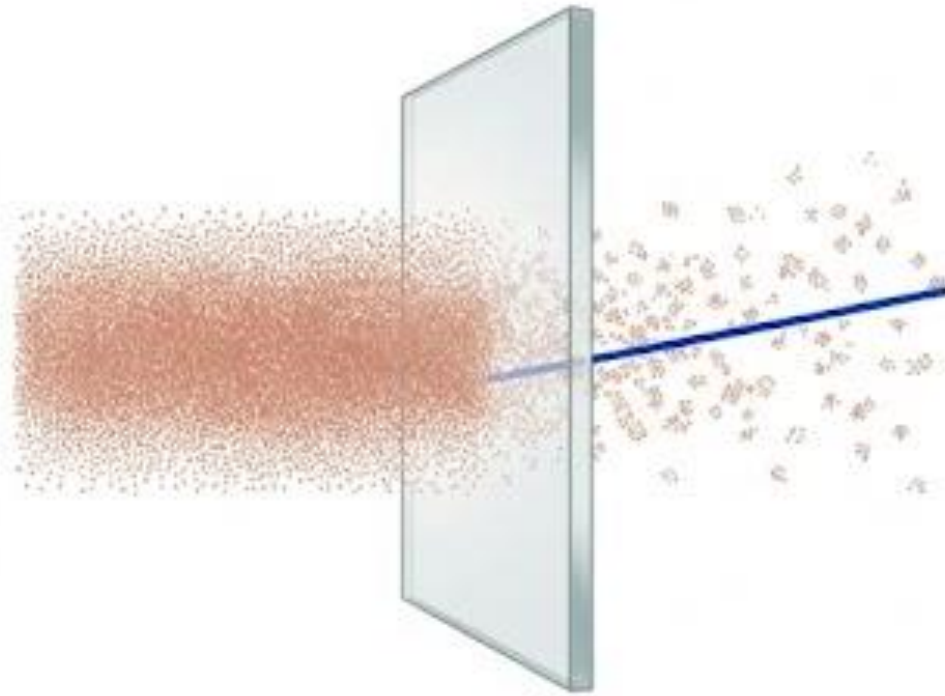
The baseline reality



- During extreme particulate events made high-humidity winter months, uncalibrated optical sensors „panicked”
- Registering false highs that severely decoupled from the gravimetric reality

Three algorithmic approaches

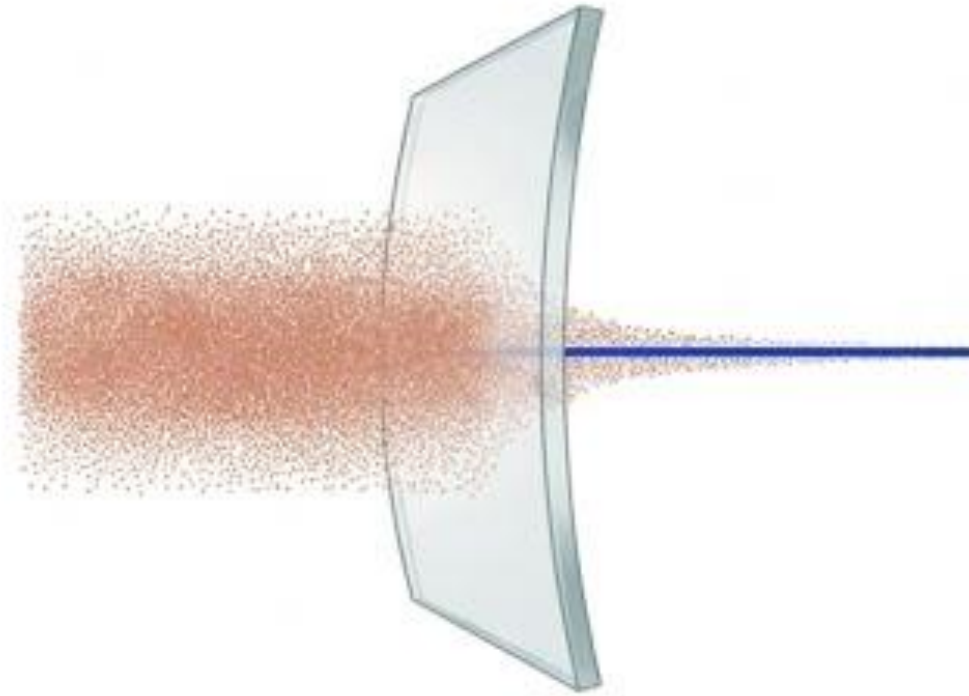
Linear Regression (LR)



Input: Single variable (PM)

Mechanism: Finds the straightest line through the noise. Simple and reliable, but completely ignores meteorological interference.

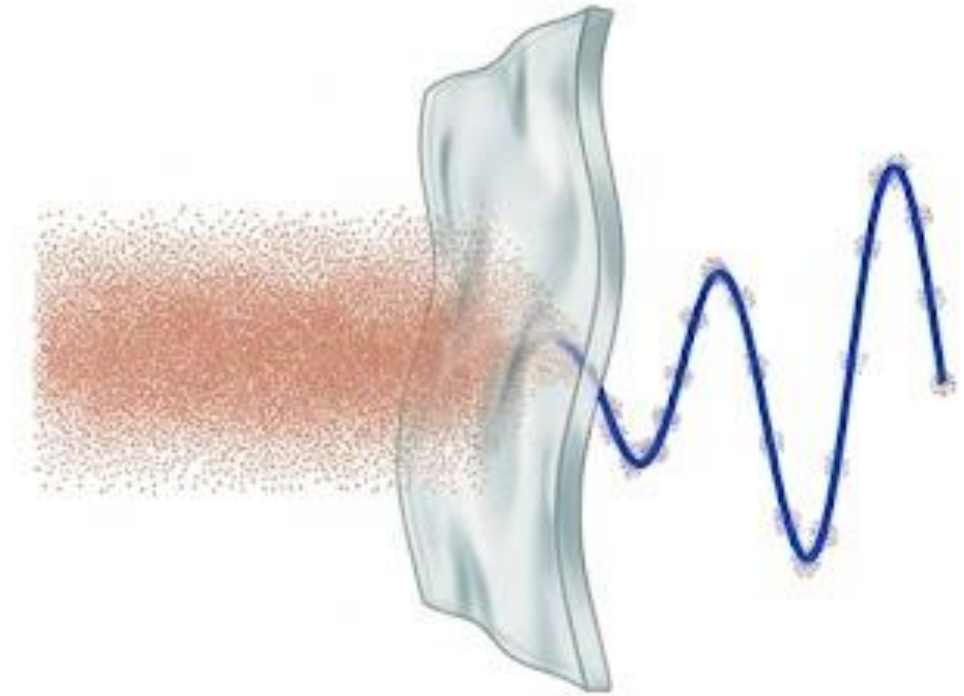
Multiple Linear Regression (MLR)



Input: Multi-variable (PM + Temp + RH + Pressure + Wind)

Mechanism: Actively accounts for the humidity trap. Complex but comprehensive; mathematically neutralizes environmental interference.

Polynomial Regression (PLR)



Input: Non-linear multi-variable

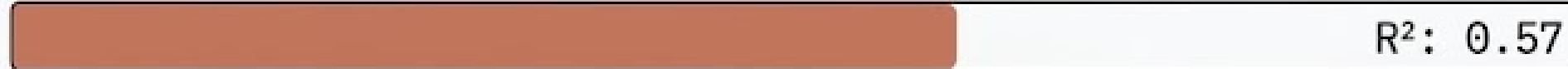
Mechanism: Bends the mathematical curve to fit the data perfectly. Highly accurate on known training data, but risks catastrophic overfitting on new horizons.

MLR tames the noise

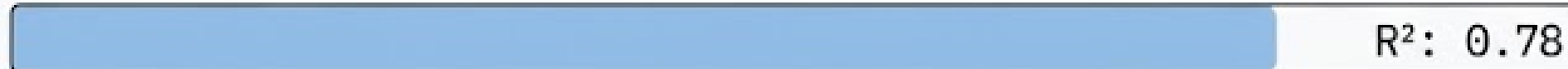
Performance Dashboard

Case Study: Sensor 2 (PM_{2.5})

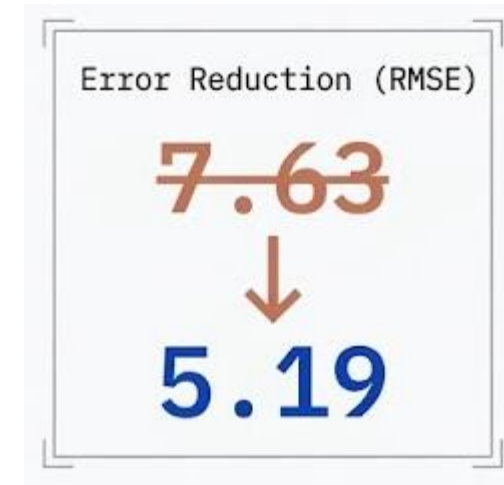
Raw Data (Uncalibrated)



LR Calibrated

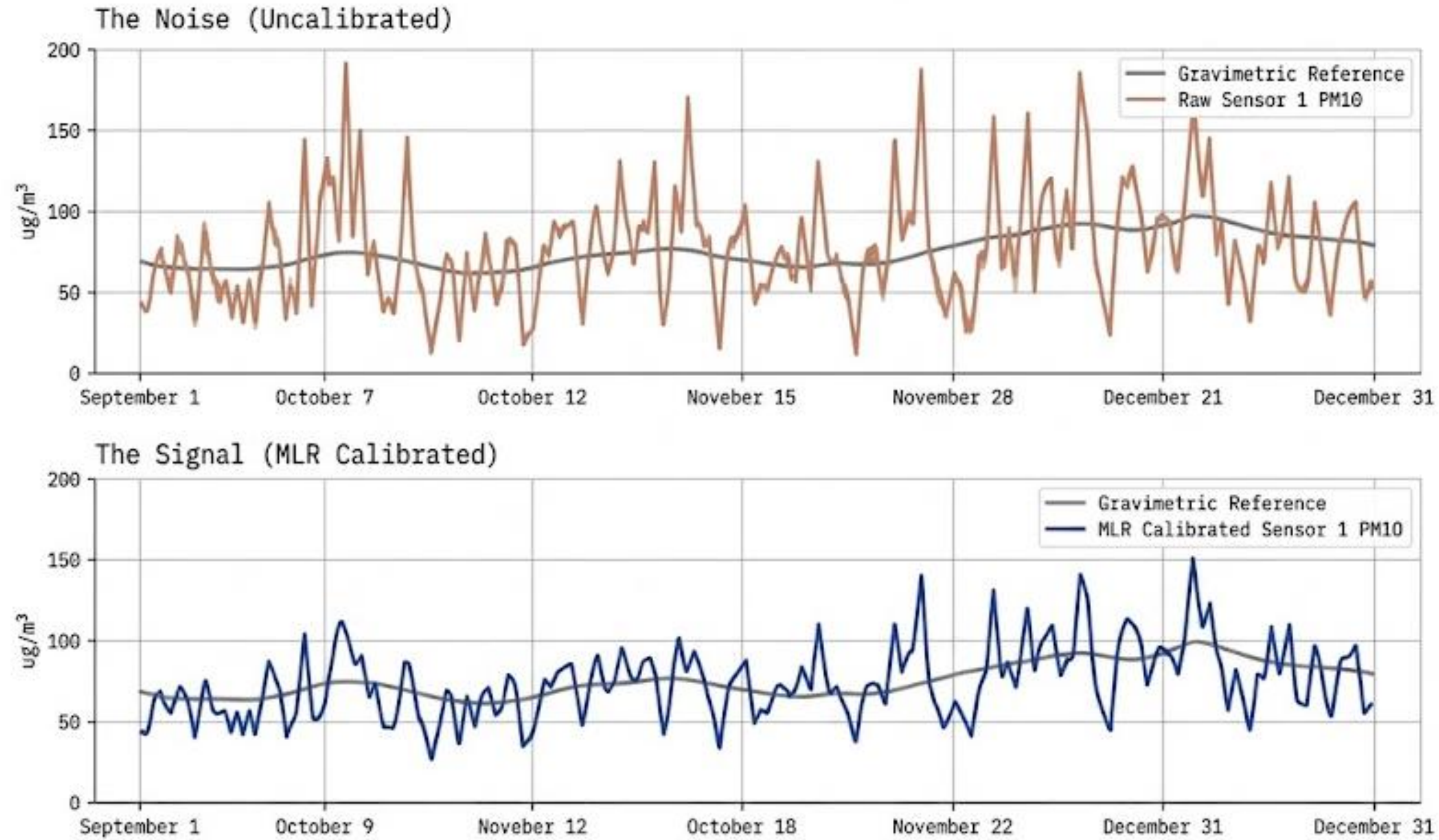


MLR Calibrated



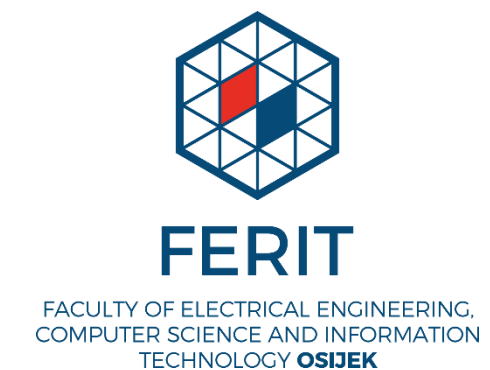
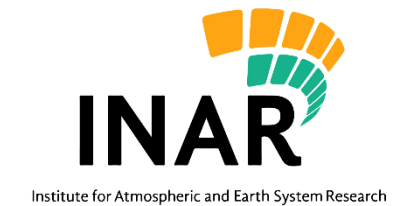
- Multiple linear regression (MLR) consistently outperforms simple linear and non-linear models for PM_{2.5} and PM₁₀ by neutralizing weather-induced bias

Visualizing the transformation





Consortium



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Thank You!

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